Grounding Industrial AI Agents with Engineering Semantics: A Layered Framework and Controlled Evaluation

Anand Todkar, Jitendra Solanki, Mrinmoy Sarkar

***Abstract*—Industrial AI agents must resolve operator requests to equipment, interpret unit-bearing values and respect access, state and process constraints. In brownfield plants these facts are distributed across controller models, engineering records and live observations; plausible answers can therefore be ungrounded and invalid actions can appear well typed. The core contribution is an evaluation of industrial information standards by their usefulness to AI agents. We introduce the agent-usability framework (AUF), organize 24 standards and vocabularies into five complementary layers, and assess how their semantics support grounding and action checking. On an authored two-line reference plant (17 assets, 73 signals, 35 tasks and 137 proposals), top-1 grounding rises from 79% with descriptions to 95% with a typed model and 100% with scoped graph retrieval. Declared-fact sufficiency rises from 23% for the typed context to 86% for the full context. A full validator detects 84/84 injected faults with no false alarms among 53 legal actions; an equally informed procedural validator matches it, showing the measured benefit is in explicit engineering facts, not SHACL itself. A separate IIH-attributed KUKA NodeSet projection retains 90 instance NodeIds and 277 references across AAS, RDF, NGSI-LD and CSV outputs, while exposing source namespace and datatype-index inconsistencies. The results provide an evidence-centered method for identifying semantic gaps and integrating retrieval with pre-action validation in industrial-agent workflows.**

***Index Terms*—Industrial agents, semantic interoperability, OPC UA, action validation, engineering knowledge graphs, programmable logic controllers.**

I. Introduction

A request to change a machine setting contains several distinct obligations: identify the machine and writable signal, interpret the requested unit, obtain current evidence, and respect operating-state and process guards. These are supervisory decisions above the controller's deterministic loop. An agent's plausible response is not itself evidence that a proposed command is admissible.

Industrial standards already describe much of this information. The problem examined here is how to expose and check those facts at agent runtime, not whether RDF is intrinsically safer than an information model or procedural implementation. OPC UA and AAS can carry richer profiles than a flat typed catalogue; our experimental conditions are selected projections, not upper bounds on the standards' capabilities [1], [2].

We pursue three contributions: (1) a capability framework linking engineering evidence to agent failures; (2) an operational integration pattern for retrieval, typed tools and pre-dispatch validation; and (3) controlled evaluation separating missing information, retrieval limitations and implementation choice. The third contribution uses controlled test cases and offline reference snapshots. PLC acquisition is distinct from validating physical process operation. Accordingly, the evaluation supports conclusions about grounding, evidence sufficiency, pre-action checks and model projection.

II. Positioning and Scope

RAMI 4.0 is an industrial reference architecture, whereas the proposed view organises the evidence needed for individual agent decisions. They are complementary: information and functional concerns provide engineering inputs; communication concerns deliver observations and commands; integration and asset concerns anchor physical identity; business concerns supply authorisation and policy. This is a conceptual crosswalk, not a normative mapping or demonstrated improvement over RAMI. The exact RAMI edition and source must be verified before submission.

The five survey layers classify identification, information models, formal semantics, domain vocabularies and cross-organisational policy. They are not a strict dependency hierarchy. Context conditions C0–C4 and validator tiers G0–G4 are experimental treatments assembled from these sources, not the survey layers themselves. Dataspace policy is not evaluated here.

Graph retrieval and knowledge-graph augmentation provide relevant prior art [3], [4]. SHACL is an existing validation language, not our invention [5]. The contribution cannot be the generic combination of an LLM and a graph. Our earlier IIH Semantics framework [6] describes Industrial Edge deployment, OPC UA aggregation, RDF/SPARQL and an automatically derived GraphQL type schema. This study evaluates evidence sufficiency, action validation and controlled ablations, rather than claiming that integration architecture as new. Specific neurosymbolic industrial systems, including the suggested CausalTrace comparator, require source and artefact verification; no capability absence or superiority is asserted before that review. The quoted 94% MAP@3 result is not used without a verified dataset, ranked-cause task and common evaluation protocol.

III. Five Layers of Engineering Semantics

The layers organise sources by their role in agent grounding, rather than replacing lifecycle or hierarchy axes in industrial reference architectures. One standard can contribute several capabilities, and a deployment can combine them without

materialising all five layers in one graph. Specifications and reference documentation provide the survey evidence; exposed implementation profiles must be assessed separately.

A. L0: Identification and property dictionaries

ECLASS, IEC common data dictionaries and industrial reference identifiers link an engineering property to a stable definition; GS1 identifiers distinguish products and individual assets across systems. This is useful when two suppliers use different labels for the same quantity, or the same label for different quantities. An agent should resolve to an identifier whose mapping is inspectable, not simply to the closest English description. Concept identity and instance identity are distinct: identifying "pressure" does not identify the pressure transmitter on a particular filler. The latter needs asset membership and a specific observation source. Persistent identifiers also require governed mappings when hardware is replaced; stability of a string alone does not establish continuity of the physical asset [7]–[10].

B. L1: Information and interface models

OPC UA combines typed address spaces, attributes, access rights and callable methods; companion models add domain content [1]. AAS supplies submodels, semantic identifiers and structured asset descriptions [2]. AutomationML supports engineering exchange, ISA-95 describes equipment hierarchy, and MTP and PackML contribute services and behaviour. WoT Thing Descriptions and NGSI-LD provide additional interface and entity representations. These technologies make introspection and tool construction possible [11]–[17], but the content actually exposed must be audited. An agent cannot check an unpublished interlock simply because the host standard is extensible.

These models can carry methods and states; the limitation is what a deployment publishes, not an inherent inability to express behaviour. Access rights differ from current-state permission, and engineering ranges differ from recipe/safety limits. Preserve each constraint's origin and version rather than collapse them into one writable flag or numeric bound.

C. L2: Formal semantics, validation and query

RDF provides globally named relations and a common graph substrate. OWL can support class and relation entailment, while SHACL checks a finite data graph against explicit shapes [5]. The distinction matters: open-world logical consistency does not by itself enforce a closed set of mandatory action parameters or establish that all interlocks have been supplied. A deployment must declare which constraints are enforced, what missing data means and which reasoning profile is supported.

SPARQL executes joins, traversals, filtering and aggregation outside the language model, returning a bounded result rather than every supporting record. PROV-O can associate transformations and observations with sources and versions. Neither a successful query nor shape conformance establishes the correctness of the engineering mapping. Graph construction, source calibration, shape completeness and unit fidelity remain upstream obligations. Declarative constraints can be shared and inspected, but a competent procedural implementation may enforce exactly the same rules; the R1 control makes that comparison explicit.

Validation lifts the proposal's target, value, unit, scope and evidence time into an action graph. Reports distinguish violations from missing evidence; bounded repair retains authorization and never weakens guards to avoid refusal.

D. L3: Quantities, observations and domain relations

QUDT supplies quantity kinds, dimensions and conversion metadata [18]. The distinction between a display symbol and a conversion definition is crucial: "bar" and "psi" name compatible units but do not themselves state the conversion coefficient. SOSA/SSN distinguishes a sensor, an observed property, an observation and its result [19]. This permits an answer to cite the measurement and acquisition time it claims to report, rather than present a number detached from a source.

Brick supplies typed equipment relations [20]; SAREF and its extensions describe devices and domain functions; DEXPI carries process-engineering structure. These should be selected for actual industrial fit rather than treated as interchangeable taxonomies. Flow connectivity, containment and observation provenance are different relations. A compressor feeding several units supports impact analysis, but a flow graph alone does not establish the causal mechanism of a recorded failure. Root-cause analysis needs incident evidence and a validated causal/diagnostic model beyond generic graph traversal.

E. L4: Cross-organisational policy and trust

Dataspaces add participant identity, catalogue metadata, exchange agreements and usage-policy mechanisms [21]. Catena-X/Manufacturing-X highlight versioned shared semantic models; ODRL expresses permissions, prohibitions and duties [22]. A supplier observation may be structurally interpretable yet unsuitable for a particular purpose or untrusted. Policies are therefore a separate gate from physical operating constraints. Policy expressions do not automatically enforce downstream confidentiality, and keeping information in an agent's memory is not universally equivalent to redistribution. Enforcement depends on the policy, processing boundary and trusted runtime. L4 is included for architectural completeness and is not experimentally validated here.

IV. Capability Framework and Operational Method

The agent-usability framework (AUF) separates identity (U1), types and access (U2), quantities and conversion (U3), relations (U4), behaviour (U5), constraints (U6), provenance and time (U7), query pushdown (U8), context economy (U9), and ecosystem/deployability (U10). These are complementary capabilities; cumulative ablation does not establish mathematical orthogonality.

The practical distinction is between facts a source can express, facts a deployment actually publishes and facts available at decision time. A unit conversion requires both source

and target definitions or a documented external conversion service. State legality requires a current state and transition model. An interlock requires its guard, dependencies and observations. Absent evidence is distinct from evidence of an illegal command.

The proposed runtime resolves the target, fetches the relevant evidence, generates a typed tool proposal and validates it. An incomplete evidence set produces abstention or a request for refresh. A proposal permitted at planning time is checked again at dispatch; the PLC retains independent, current-state checks and final actuator authority. A finite retry budget, operator approval and an audit trail are required for repair. No LLM participates in a hard-real-time loop.

AUF scores remain expert assessments until independently checked. The planned assessment uses two scorers, specification-clause evidence, ordinal agreement and adjudication, then tests whether AUF-guided capability selection improves a held-out installation within a fixed integration budget. Those results are not available in this revision.

A. Assessment rubric and operational failure mapping

The ten capabilities were selected by tracing the minimum evidence needs of a bounded industrial-agent cycle, then adding its deployment constraints. U1–U8 cover resolving the requested asset (identity), interpreting its interface (types/access), interpreting values (units), locating it in plant context (relations), determining available operations (behaviour), checking proposed actions (constraints), establishing whether observations are current and traceable (provenance), and obtaining relevant facts efficiently (query). Each addresses a distinct failure class in the benchmark taxonomy. U9 and U10 are cross-cutting feasibility criteria: whether a faithful evidence slice fits the agent's context budget, and whether implementations, converters and governance support deployment. Thus AUF is a coverage-oriented checklist for industrial grounding and supervisory action, not a claim to enumerate every need of every agent or prove mathematical completeness. Some criteria interact; none dictates a preferred serialization. Procedural code may provide the same facts and checks as a semantic technology.

Table I retains 24 assessments. Scores distinguish absent, partial/indirect, explicit/incomplete and strongest criterion-specific support; they are ordinal, not an accuracy predictor. Versions and exposed profiles can change judgements. Widespread typing but concentrated unit/constraint support motivates selective composition, not adoption of every technology.

B. An evidence contract for each proposal

Let F denote available engineering facts and $R(q)$ the declared evidence requirements of task q. The reported coverage and sufficiency are

$$c(q,F)=\frac{|R(q)\cap F|}{|R(q)|},\quad s(q,F)=\mathbf{1}[R(q)\subseteq F]. \tag{1}$$

These quantify a particular evidence contract; another valid derivation may need a different requirement set. For an action, validation has three dispositions: reject if a violation is proved, abstain if required evidence is unresolved, and accept only relative to the checked contract if neither condition applies. Acceptance is not a proof that all physical hazards have been modelled.

For compatible affine units, a proposed source value x is converted by

$$x_t=\frac{m_s x+b_s-b_t}{m_t}, \tag{2}$$

with dimension compatibility checked first. Engineering bounds apply to x_t, not the source numeral. A faithful numeric answer additionally compares x_t with a referenced observation under a declared tolerance. A writable point's guard predicates and operating state are separate from that numeric check.

C. Runtime insertion points and engineering ownership

Semantics enters retrieval, tool-schema synthesis, constrained decoding and post-proposal verification. Retrieval resolves equipment before projecting relevant values, relationships and guards. Tool schemas can constrain identifiers, enumerations and types; only decoder-enforced features eliminate calls at generation time. Snapshot-dependent conditions still require live validation. A repair report should name the violated rule and missing evidence; retries remain bounded and authorization unchanged.

The source model stays behind a query interface. A parameterised service can answer a plant-wide aggregate, expose current admissible commands or return an evidence slice with provenance. MCP is one possible transport for such tools, not a source of equipment truth. Integration must authenticate callers, constrain scope, preserve mapping versions and avoid unrestricted graph or PLC access. Figure 1 separates supervisory proposal generation from controller-side execution authority.

V. Controlled Evaluation

The stored reference-plant model contains 17 assets and 73 signals. Its 35 tasks and 137 proposed actions are retained for comparability, not presented as independently collected industrial workloads. All revision outputs are isolated from the original manuscript and archived results.

A. PLC acquisition and cross-format transformation

The authors report acquisition on 2026-10-08 using Siemens IIH 2.6.1 on an Industrial Edge IPC427E; the product reference is the v2.6 documentation [23], distinct from the installed patch version. This aggregation server connects as an OPC UA client to the S7-1500 CPU 1518-4 PN/DP server, reading test variables rather than physical I/O. The workflow is PLC $\rightarrow$ IIH $\rightarrow$ offline snapshot and NodeSet2, AAS JSON, RDF, JSON-LD, WoT TD, NGSI-LD and CSV projections.

The PLC is the source; IIH provides an aggregated view. A preservation audit must map source/IIH identifiers and compare types, references, units, access and timestamps rather than assume identical models. Exports, downstream extraction and linkage to stored declarations remain undocumented. Added

TABLE I

RETAINED AUF EXPERT ASSESSMENT FROM THE ORIGINAL SURVEY, USING CRITERION-SPECIFIC SCORES 0–3. THESE ARE SPECIFICATION ASSESSMENTS, NOT MEASURED AGENT PERFORMANCE OR INDEPENDENTLY VALIDATED RANKINGS. U1 IDENTITY; U2 TYPES; U3 UNITS; U4 RELATIONS; U5 BEHAVIOUR; U6 CONSTRAINTS; U7 PROVENANCE; U8 QUERY; U9 ECONOMY; U10 ECOSYSTEM.

Layer	Technology	U1	U2	U3	U4	U5	U6	U7	U8	U9	U10
L0	ECLASS / IEC CDD / IEC 61987	3	2	2	1	0	1	0	1	2	3
L0	GS1 identifiers / EPCIS	3	1	0	1	1	0	2	1	2	3
L1	OPC UA core and companions	3	3	2	2	3	1	2	1	1	3
L1	Asset Administration Shell	3	3	2	2	2	1	2	1	1	3
L1	AutomationML	2	3	1	3	1	1	1	0	1	2
L1	MTP	2	3	2	2	3	2	1	0	2	2
L1	PackML	1	2	0	0	3	2	0	0	3	3
L1	ISA-95 / IEC 62264	2	2	1	3	1	0	1	0	2	3
L1	W3C WoT TD 1.1	3	3	2	1	3	1	1	1	2	2
L1	NGSI-LD	3	2	2	3	1	1	3	2	2	2
L1	IEC 61850	3	3	2	2	2	1	2	0	1	3
L2	RDF / RDFS	3	2	0	3	0	0	1	3	1	3
L2	OWL 2	3	3	0	3	0	2	0	2	1	3
L2	SHACL	2	3	1	2	1	3	1	2	2	3
L2	SPARQL 1.1	2	1	0	3	0	1	1	3	3	3
L2	PROV-O	3	2	0	3	1	0	3	2	2	2
L3	QUDT	3	2	3	1	0	2	0	2	2	2
L3	SOSA / SSN	3	2	1	2	1	0	3	2	2	3
L3	Brick	3	3	1	3	0	1	0	3	2	3
L3	SAREF and extensions	3	2	2	2	2	0	1	2	2	2
L3	DEXPI	2	3	2	3	0	1	1	1	1	2
L4	IDS-RAM / Dataspace Protocol	3	2	0	2	1	2	3	1	1	2
L4	Catena-X / Manufacturing-X	3	3	1	2	1	2	3	1	1	2
L4	ODRL	2	2	0	2	2	3	2	1	2	2

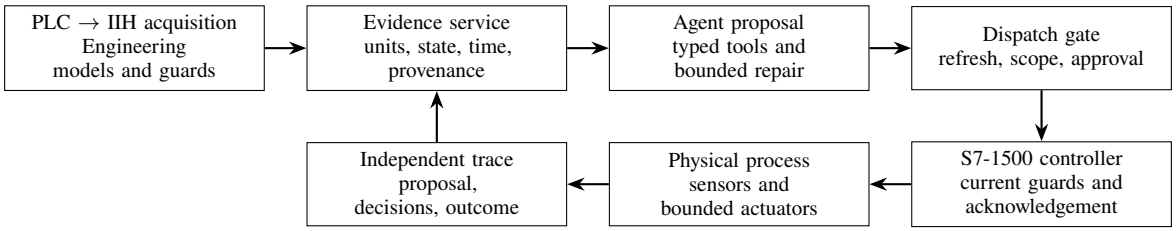


Fig. 1. Proposed supervisory architecture. Evidence supports the agent; dispatch and PLC checks retain current-condition authority. This is a design, not evidence of a completed physical trial.

engineering semantics require separate provenance. Archived results remain unchanged; E11 dated 2026-10-06 cannot originate from this capture. Acquisition alone does not validate physical process operation.

B. Ad hoc KUKA NodeSet projection check

Separately from the generated bottling benchmark, the supplied KUKA instance NodeSet and Robotics/DI companion models were projected into AAS, RDF/JSON-LD, NGSI-LD and CSV. All 90 source instance NodeIds and 277 references were retained in each projection; the RDF syntaxes round-trip isomorphically. The source namespace mismatch and six datatype-index conflicts are reported, not repaired. This structural check is not fed into E1–E11 and supports no task-accuracy, physical-operation or external AAS/NGSI-LD conformance claim; byte costs and full audit are in the ad hoc artifact bundle.

C. Original context conditions and task corpus

The two-line beverage plant combines 22 shared descriptions, metric/imperial units, PV/SP pairs, four interlocks and 14 feed relations. C0 exposes tags and values; C1 adds descriptions/unit symbols; C2 adds quantity kinds, ranges, alarms, access, roles, membership, parentage and selected semantic identifiers. C3 adds unit definitions/conversions, topology, taxonomy and location. C4 adds states/legal commands, guards, observation time, events and recipes. These are selected profiles, not limitations inherent to their host standards.

Tasks cover grounding (GRD), units (UNI), ranges (RNG), topology (TOP), provenance (PRV), action legality (ACT), guards (SAF) and recipes (REC); 19 identify a single gold tag. The proposed-action corpus has 84 faulty and 53 legal controls across 12 injected families. G0 accepts everything; G1 checks signatures; G2 adds references/access/ranges and literal units; G3 adds unit/relational checks; G4 adds behaviour, guards and observations. Original lexical versus graph retrieval changes jointly with the conditions, a confound partially isolated by R2. The 6,000-token default and six-point budget sweep use the surrogate tokenizer. Every task declares required fact keys, making omissions inspectable, but the authors' designed taxonomy limits external validity.

D. R1: Equal-information validation

Both implementations receive the same engineering ranges, unit calculus, asset containment, PackML state/commands, interlocks and observations. The procedural implementation reads those declarative reference data without calling SHACL. Predictions do not inspect gold labels. Raw constraint-family sets are compared; diagnosis maps an injected scale error to a range violation identically for both implementations, rather than adding labels to predictions from gold. Detection, false positives and diagnosis are reported separately. Binomial Wilson intervals are archived but are descriptive for this designed corpus, not evidence of installation-wide incident prevalence.

E. R2: Retrieval crossed with context

Lexical retrieval is evaluated under all conditions. Membership-scoped retrieval is added where published asset membership is available; both scoped strategies use a fixed manually supplied entity resolver. Graph traversal is restricted to C3/C4. Each supported strategy is tested at 1,500, 6,000 and 24,000 surrogate tokens with a fixed condition renderer. An unbudgeted all-elements context is an oracle diagnostic, not a deployable retriever. No required-fact set is used for ranking. This matrix separates some retrieval effects but does not equalise all notation familiarity or entity-linking assumptions.

F. R3–R5: Ablation, degradation and fidelity

R3 removes units, flow topology, behaviour, constraints or provenance separately. The context ordering remains fixed to full C4 as a diagnostic intervention; it is not evidence that an ablated deployment could build that ordering. The validator separately removes access to relevant checking capabilities and counts abstentions, which must not be counted as detections. Context flow topology and validator containment are different interventions and cannot be pooled as one effect size.

R4 probes missing state, unknown/missing unit definitions, malformed timestamps, missing observations and a synthetic planning-to-dispatch state change. R5 serialises one RDF graph and parses each syntax back, qualifying an equal-information comparison only if graph isomorphism is preserved.

Evidence sufficiency is the fraction of tasks whose declared required facts are present. It is not an unconditional accuracy bound: alternative derivations, external knowledge and inferred conventions may change answerability. Conversion facts are conservatively counted only when both unit definitions occur in the context. Token counts are a deterministic surrogate, not model billing tokens.

VI. Original Benchmark Results

A. E1, E2 and E6: Resolution and evidence sufficiency

Table II distinguishes locating a likely tag from carrying enough facts for a checkable answer. C1 resolves 79% of single-tag questions but provides no explicit membership proof; C2 reaches 95% top-1 with 23% task sufficiency. C3/C4 achieve perfect top-1 resolution under their scoped resolver, yet context can omit a fact required for audit. Verifiable grounding checks the membership evidence, not merely whether the ranked identifier equals the gold identifier.

Table III localises the archived treatment effects. Typing supports identity and many range questions; unit/topology facts support quantity reporting and impact tasks; state, guards and provenance support the remaining operational categories. These categories were constructed around the capabilities, so the block structure is evidence within the benchmark, not a proof that every future task has one uniquely necessary layer.

Increasing the budget from 750 to 24,000 tokens raises C2's archived sufficiency from 0.229 to 0.257, versus 0.171 to 0.971 for C4. More space cannot reveal fact kinds omitted by the selected projection, but can help a richer projection retain them. This does not establish that a more complete OPC UA/AAS profile would show the same plateau. At fixed budgets extra state, event and guard records can displace measurements, motivating task-dependent retrieval.

B. E3: Format costs and context economy

The archived CSV costs 2,256 surrogate tokens, OPC UA XML 52,324 and AAS JSON 65,929 (Table IV); N-Triples is substantially larger. These figures describe emitted artefacts, not an intrinsic ranking of standards: CSV carries fewer facts, projections differ and formatting affects the count. An exchange format can be useful even if it is poor prompt material. The engineering implication is to serve typed queries or selected evidence slices, while R5 checks whether syntax-only comparisons preserve the graph. Billing claims require the actual model tokenizer and repeated tool-schema overhead.

C. E4 and E4b: Faults, false alarms and procedural checks

G2 detects 57 of 84 faulty proposals, but rejects four legal unit re-expressions. G3 detects 62 without those false alarms, and G4 detects all 84 (Table V). Literal unit comparison explains both unnecessary rejection and incorrect diagnosis of scale errors. The family breakdown (Table VI) includes invalid references, arguments/enumerations, access, ranges, dimension/scale, scope, guards, states, staleness and unfaithful values. Behaviour/provenance families total 22 faults in this designed corpus, not a frequency estimate for real incidents.

The original GP baseline supplies unit calculus but not full behaviour, guard or observation models (Table VII). It removes unit false alarms but cannot show semantic validation outperforming equally informed procedural rules. R1 supplies that comparison. Archived diagnosis also identifies the injected scale family through gold-dependent reclassification; the revised raw-family audit avoids that prediction-time label dependence.

D. E5: Admissible calls and schema cost

The reference enumerates 71,540 well-formed write triples and 153 command calls. Only 710 writes and 26 commands are legal in its current state. G2 admits 190 writes because literal-unit restrictions exclude many legal re-expressions; G3 admits 755 and G4 the 710 reference-legal writes. Table VIII reports admitted-set precision and recall. Uniform-call legality

TABLE II

E1/E2/E6: ARCHIVED GROUNDING AND EVIDENCE SUFFICIENCY AT 6,000 SURROGATE TOKENS. SUFFICIENCY USES THE ORIGINAL DECLARED-FACT BOOKKEEPING, NOT A UNIVERSAL ACCURACY BOUND.

Context	Top-1	MRR	R@5	Verif.	Distr@5	Coverage	Suff.	Tokens	Cov./1k
C0	0.000	0.000	0.000	0.000	0	0.100	0.000	407	0.247
C1	0.789	0.866	1.000	0.000	10	0.252	0.000	1310	0.192
C2	0.947	0.974	1.000	0.947	7	0.549	0.229	5193	0.106
C3	1.000	1.000	1.000	1.000	4	0.761	0.514	5332	0.143
C4	1.000	1.000	1.000	0.842	4	0.857	0.857	5290	0.162

TABLE III

E6: ARCHIVED EVIDENCE SUFFICIENCY BY TASK CATEGORY. GRD IDENTITY; UNI UNITS; RNG RANGES; TOP TOPOLOGY; PRV PROVENANCE; ACT ACTIONS; SAF GUARDS; REC RECIPES.

Context	GRD	UNI	RNG	TOP	PRV	ACT	SAF	REC
C0	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
C1	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
C2	1.000	0.000	0.600	0.000	0.000	0.000	0.000	0.000
C3	1.000	0.800	0.800	1.000	0.000	0.000	0.000	0.000
C4	0.800	0.800	0.600	1.000	0.750	1.000	1.000	1.000

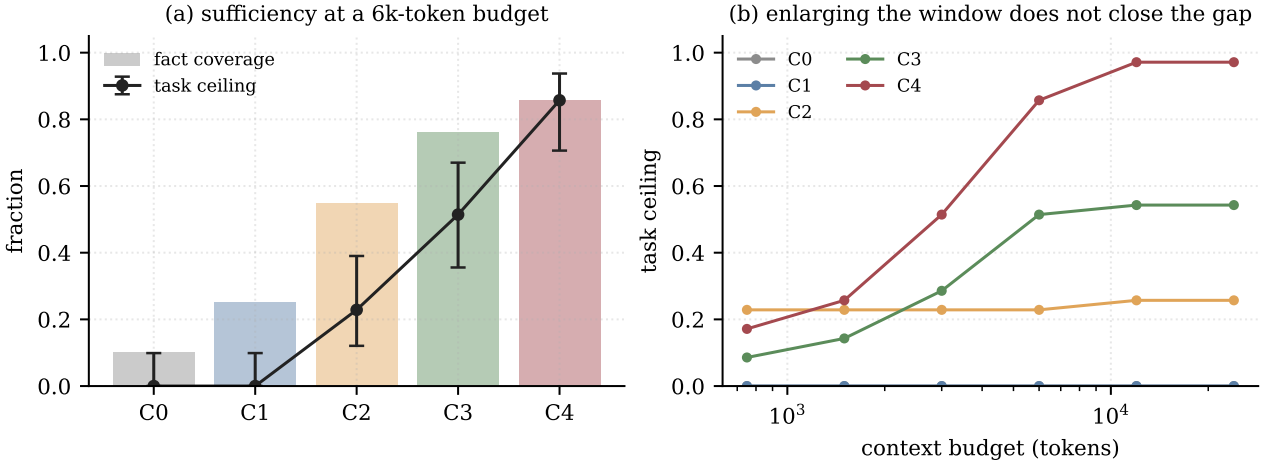


Fig. 2. Retained E2/E6 figure. The original axis term "ceiling" means declared-evidence sufficiency, not an unconditional LLM accuracy bound. Original conversion bookkeeping is preserved; audited R2 results are separate.

TABLE IV

E3: ARCHIVED FORMAT COSTS. FORMATS DO NOT PRESERVE IDENTICAL FACT INVENTORIES; RATIOS MIX CONTENT AND SYNTAX. R5 TESTS FIDELITY SEPARATELY.

Format	Tokens	Ratio/CSV
CSV	2256	1.0
NGSI-LD	11233	5.0
WoT TD	18471	8.2
Turtle	45803	20.3
OPC UA XML	52324	23.2
AAS JSON	65929	29.2
N-Triples	179241	79.5
JSON-LD	145260	64.4

TABLE V

E4: ARCHIVED VALIDATOR RESULTS ON 84 FAULTY AND 53 VALID ACTIONS. INTERVALS ARE DESCRIPTIVE WILSON RECALL INTERVALS.

Tier	Caught	Recall	95% CI	Diag.	FPR
G0	0/84	0.000	[0.00,0.04]	0.000	0.000
G1	12/84	0.143	[0.08,0.23]	0.143	0.000
G2	57/84	0.679	[0.57,0.77]	0.631	0.075
G3	62/84	0.738	[0.64,0.82]	0.738	0.000
G4	84/84	1.000	[0.96,1.00]	1.000	0.000

is a property of this enumeration, not an estimate of an LLM's proposal distribution.

The write schema grows from 205 surrogate tokens at C0 to 981 at C2, 1,669 at C3 and 2,317 at C4. Static enums and numeric limits can constrain generation, but live-state facts in a cached schema expire. Cost is not assumed to be paid only once: models may receive tools repeatedly, and refresh policies must be documented. Decoder support and later validation jointly determine which calls are impossible versus merely checked after generation.

TABLE VI
E4: DETECTED COUNTS BY INJECTED FAULT FAMILY. R1 AUDITS RAW FAMILY SETS WITHOUT GOLD-DERIVED RECLASSIFICATION.

Family	n	G1	G2	G3	G4
F-ACC	8	0	8	8	8
F-CARD	8	8	8	8	8
F-DIM	13	0	13	13	13
F-ENUM	4	4	4	4	4
F-ILK	3	0	0	0	3
F-RANGE	8	0	8	8	8
F-REF	12	0	12	12	12
F-SCOPE	5	0	0	5	5
F-STALE	4	0	0	0	4
F-STATE	7	0	0	0	7
F-UNITSCALE	4	0	4	4	4
F-VALUE	8	0	0	0	8

TABLE VII
E4B: ORIGINAL UNIT-CALCULUS PROCEDURAL BASELINE (GP), NOT THE EQUALLY INFORMED R1 IMPLEMENTATION.

Tier	Recall	FPR	Scale	Behav.
G2	0.679	0.075	0/4	0/22
GP	0.679	0.000	4/4	0/22
G3	0.738	0.000	4/4	0/22
G4	1.000	0.000	4/4	22/22

E. E7: Query execution instead of stuffing

Eight hand-authored queries cost 79–287 tokens including results, against 741–5,501 for stuffed C4 contexts (Table IX). Savings range from about 3.3 to 69 times in the saved JSON. RNG-04 checks plant-wide alarm violations; its top-k context is insufficient at all tested budgets, while a query returns the needed aggregate. This concerns execution of an existing correct query, not query generation, repairs, endpoint latency or access control. Parameterised reviewed queries can expose this benefit without arbitrary graph access.

F. E8 and E9: Naming and projection robustness

Re-tagging changes identifiers but holds engineering facts fixed. Mnemonic decoding reaches 95% grounding on its native convention but varies by 53 percentage points across schemes. With misleading mnemonics it reaches 63% against 84% without decoding. Explicit membership keeps the typed resolver at 95% and the graph resolver at 100% for these perturbations (Table X). The original "confidently wrong" field means a non-empty wrong resolution, not calibrated model confidence.

Original routed C4 projection raises declared sufficiency from 0.857 to 0.943 and verifiable grounding from 0.842 to 1.000 at nearly equal token cost (Table XI). The router is a fixed lexicon, not a learned policy or universal lower bound. R2's conversion audit changes sufficient-task sets; the two findings must not be pooled as though the evidence definitions were identical.

VII. Revision Controls and Interpretation

R1 produces 0 raw-family disagreements. Both implementations detect every injected fault and accept every legal reference action (Table XII). This is a negative result for an engine-superiority claim and positive evidence that the supplied facts can be checked independently. Standardisation, reuse and maintenance advantages need separate effort measurements.

Table XIII distinguishes restricted contexts from unbudgeted evidence. Under the audited conversion definition, a task may lack an explicit target-unit definition even when the original bookkeeping marked conversion as available. Differences from archived results are therefore not silently treated as replication failures or attributed solely to routing.

Removal effects depend on the workload and budget (Table XIV). Smaller contexts can offset lost capability for some tasks; lack of a net sufficiency decrease does not prove a capability unnecessary. Additional per-category and per-task outputs are archived. The corpus was designed around these failure families, so general necessity requires independently authored tasks and external workloads.

The synthetic state probe permits a volume change in Idle and rejects it after a change to Execute. It demonstrates the test harness's revalidation rule, not that a real PLC/agent pipeline prevents a timing race. Missing evidence remains visible in the verdict even when a separately proven fault also requires rejection.

The installed RDFLib Turtle serializer rounds some double literals, causing a strict graph mismatch despite identical triple counts. Such output is excluded from equal-information claims; N-Triples and JSON-LD qualify in this run. Preserving datatype and numeric precision is a deployment requirement, not merely a token-economy detail.

VIII. GPT Evaluation and E11 Provenance

The original generated E11 report names the Azure deployment `gpt-5.6-sol`, dated 2026-10-06, with 35 tasks per condition and three repeats. Its retained aggregates rise from zero to 55% accuracy and from 100% to 28% refusal (Table XVII). At C2 the reported 29% accuracy exceeds 23% sufficiency, illustrating why sufficiency is not an unconditional accuracy bound. Reported conditional accuracy decreases from 96% at C2 to 64% at C4. Naming conventions and model priors are possible explanations, not causally identified mechanisms in this run.

However, the only retained raw JSON, including its committed historical version, contains GRD-01 with one repeat in each condition: five responses, not the 525 implied by the full-suite report. Table XVIII is generated from those rows and matches the archive's own counts. The smoke test refuses at C0 and answers that grounding task correctly at C1–C4. It cannot corroborate behavioural, safety, recipe or provenance categories, nor establish full-suite accuracy. The retained aggregate table is included transparently but cannot currently be reconstructed; reconciliation is required before submission.

The model string is a recorded deployment identifier, not independently verified public model lineage. Only one archive exists; no plural-model comparison is claimed. Heuristic grading and sufficiency labels require manual audit for any recovered full run. New calls would be a new dated evaluation, not

TABLE VIII
E5: admitted write and command spaces; P and R are precision and recall against reference legality.

Tier	Writes	Write P	Write R	Bits removed	Commands	Cmd P	Cmd R
G0	71540	0.010	1.000	0.00	153	0.170	1.000
G1	71540	0.010	1.000	0.00	153	0.170	1.000
G2	190	0.937	0.251	8.56	81	0.321	1.000
G3	755	0.940	1.000	6.57	81	0.321	1.000
G4	710	1.000	1.000	6.65	26	1.000	1.000

TABLE IX
E7: hand-authored SPARQL execution costs, including query and result. Suff. marks the original stuffed context; query generation is not evaluated.

Task	Query+result	Stuffed	Ratio	Suff.
UNI-05	153	5373	35.1	yes
RNG-04	222	741	3.3	no
TOP-02	79	5461	69.1	yes
TOP-03	139	5172	37.2	yes
ACT-04	84	5172	61.6	yes
SAF-04	241	5172	21.5	yes
PRV-04	287	5492	19.1	yes
GRD-01	149	5501	36.9	yes

TABLE X
E8: top-1 accuracy / wrong non-empty resolution across naming conventions. Wrong non-empty resolution is not calibrated model confidence.

Scheme	C1	C1+decode	C2+decode	C4
mnemonic	0.79/0.21	0.95/0.05	0.95/0.05	1.00/0.00
opaque	0.53/0.47	0.53/0.47	0.95/0.05	1.00/0.00
kks	0.89/0.11	0.89/0.11	0.95/0.05	1.00/0.00
vendor	0.42/0.58	0.42/0.58	0.95/0.05	1.00/0.00
inconsistent	0.84/0.16	0.63/0.37	0.95/0.05	1.00/0.00

TABLE XI
E9: original neighbourhood (N) and routed (R) projection at 6,000 tokens. R2 uses a more conservative conversion-evidence audit.

Context	Mode	Suff.	Coverage	Verif.	Tokens
C3	N	0.514	0.761	1.000	5332
C3	R	0.543	0.774	1.000	5341
C4	N	0.857	0.857	0.842	5290
C4	R	0.943	0.959	1.000	5345

TABLE XII
R1: equal-information validation on the synthetic reference corpus. Diagnosis maps scale faults to range violations for both implementations.

Validator	Recall	FPR	Diagnosis
Procedural	1.000	0.000	1.000
SHACL	1.000	0.000	1.000

recovery. Prompts, responses, sampling/tool settings, grader version and evidence definition must be frozen together; repeats are not independent industrial observations.

TABLE XIII
R2: crossed retrieval at 6,000 surrogate tokens. Oracle-all is unbudgeted and diagnostic only. Membership uses a fixed shared entity resolver.

Context	Retrieval	Suff.	Coverage
C2	lexical	0.229	0.549
C2	oracle_all	0.314	0.649
C2	membership	0.229	0.485
C4	lexical	0.714	0.846
C4	oracle_all	0.971	0.995
C4	membership	0.457	0.688
C4	graph	0.800	0.929

TABLE XIV
R3: capability removal. Context sufficiency uses a fixed full-C4 ordering; abstentions are not detections. Context topology removes flow only; validator topology removes scope-containment checks.

Removed	Suff.	Recall	Abstain
none	0.800	1.000	0
behaviour	0.629	0.893	37
constraints	0.686	0.964	20
provenance	0.743	0.845	40
topology	0.629	0.940	5
units	0.800	0.607	60

TABLE XV
R4: synthetic missing-evidence and dispatch-time state probes. These are not PLC trials or physical safety measurements.

Probe	Disposition
before_state_change	accept
after_state_change	reject
missing_state	abstain
missing_unit	reject
unknown_unit	reject
malformed_time	reject
missing_observation	abstain

TABLE XVI
R5: identical RDF graph round-tripped through three syntaxes. Only isomorphic round trips qualify as equal-information controls. Counts use the surrogate tokenizer.

Syntax	Triples	Tokens	Isomorphic
turtle	2465	45803	no
nt	2465	179432	yes
json-ld	2465	145326	yes

TABLE XVII
ARCHIVED E11 GPT DEPLOYMENT EVALUATION: `GPT-5.6-SOL`, ACCESSED 2026-10-06; 35 TASKS PER CONDITION AND THREE REPEATS. VALUES ARE ROUNDED FROM THE ARCHIVED REPORT. THIS IS A MODEL EVALUATION ON SYNTHETIC PLANT CONTEXT, NOT A HARDWARE TRIAL.

Context	Accuracy	Suff.	Refusal	Wrong/ins.	Acc./suff.
C0	0.00	0.00	1.00	0.00	–
C1	0.14	0.00	0.85	0.01	–
C2	0.29	0.23	0.63	0.10	0.96
C3	0.40	0.51	0.51	0.00	0.78
C4	0.55	0.86	0.28	0.00	0.64

TABLE XVIII
E11 RAW-DATA AUDIT: CURRENTLY RETAINED GPT SMOKE-TEST RESPONSES. ONLY GRD-01 AND ONE REPEAT ARE AVAILABLE; NO FULL-SUITE CONCLUSION FOLLOWS.

Context	Responses	Accuracy	Refusal	Suff.
C0	1	0.000	1.000	0.000
C1	1	1.000	0.000	0.000
C2	1	1.000	0.000	0.229
C3	1	1.000	0.000	0.514
C4	1	1.000	0.000	0.857

IX. HARDWARE-BASED EVALUATION DESIGN

A. Controller acquisition and physical validation boundary

IIH 2.6.1 on an Industrial Edge IPC427E collects the CPU 1518-4 PN/DP test-variable model. This author-confirmed 2026-10-08 acquisition has no attached source/aggregate exports or mapping audit. R4 remains an offline probe; no physical sensor/actuator trial logs are available. The scenarios below are proposed physical validation, not measured acquisition outcomes.

The recommended rig is two guarded 24 V handling/indexing stations, optionally coupled to a low-head water-filling module. Real presence sensors and physical counters support identity/state tasks; speed feedback or an encoder is needed if actual speed tracking is assessed. A reservoir, bounded pump/valve and catch vessel, with level switches, flow sensing or a calibrated scale, support unit-correct fill-volume tasks. Avoid adding a pressure or heater hazard solely to imitate the generated plant. Final equipment and permitted disturbances require a competent engineer's local risk assessment.

In the proposed execution study, the S7-1500 retains sequencing, operating guards, output authority and acknowledgement. The acquisition IPC427E could also host the graph/SHACL services, procedural baseline, supervisor and historian after resource checks. IPC227G can provide separate replay/disturbance services; its exact order number and resources must be checked. LLM inference can be remote or on a suitable additional host; the IPCs are not presumed capable of running a large model locally. Record CPU order number, firmware, TIA version, I/O, OPC UA support/licence, IPC OS, RAM and network interfaces. A standard CPU is not presumed safety-rated.

TABLE XIX
PLANNED HARDWARE SCENARIOS. NONE HAS A MEASURED OUTCOME YET. INVALID PROPOSALS ARE ASSESSED IN SHADOW MODE, NOT SENT TO ACTUATORS.

Scenario	Required observable outcome
Identity/access	Correct Station A/B source; PV/SP distinction; no PV write.
Units/fill	Converted value with declared units, sensor tolerance and measured volume.
State/guard	Legal command and vessel/level guard checked against current PLC state.
Stale/reconnect	Observation age, quality, refresh/abstention and reconnect trace.
Dispatch race	State change between proposal and commit; current-state rejection.
Scope/topology	A feeds B; unavailable B or restricted scope reflected in decision.
Incident cause	Engineer-labelled disturbance, evidence and ranked cause candidates.
Transfer	Held-out station/configuration; edits, regression failures and engineer-hours.

B. Command interface and current-state protection

The agent receives allowlisted supervisory tools, not raw outputs or unrestricted memory/programming access. Publish immutable asset IDs, source NodeIds, units, access, bounds, current state, observation time/quality and mapping versions. Use similar DisplayNames on Stations A/B but unique identities so that grounding errors are observable without compromising actual control.

A command mailbox should carry a unique request ID, target, opcode, native-unit value, approval reference and expiry/state reference. Multi-field OPC UA writes are not presumed atomic: stage parameters and use a sequence/commit handshake; the PLC reads a consistent request before execution. It independently checks current guards and states, rejects expired/duplicate or unauthorised requests, and records acceptance/rejection reason and acknowledgement ID. A proposal valid in Idle must not be executed after a transition to Execute merely because an earlier graph snapshot permitted it.

Commission duplication, reconnect and timeout behaviour without an LLM first. If evidence is stale, quality bad, mapping inconsistent or acknowledgement missing, refresh or abstain rather than issuing an uncertain retry. PLC-side guards and appropriate independent stop/protection remain authoritative if the graph, IPC, model or network fails. OPC UA security configuration and accounts follow local engineering practice; credentials never enter trial logs. The agent remains outside hard-real-time control loops.

C. Trial sequence and matched comparisons

Start with manual/controller-only commissioning, calibration and protection verification. Freeze an independently reviewed task list, operating envelope, labels and episode split before running models. In read-only shadow mode, compare proposed actions over the same live snapshots. Never execute a known invalid action to demonstrate rejection. Approved legal, bounded proposals can then be enabled under supervision, with independent sensor outcome logging. Unsafe disturbances use replay and retain replay provenance rather than being reported as physical incidents.

Compare a realistic typed OPC UA/AAS baseline, equally informed structured-record retrieval, full procedural validation and declarative integration. Preserve model, prompts, permissions, snapshots and approval policy within paired comparisons. The typed baseline must retain behaviour it actually exposes. Hold out Station B or a changed recipe/configuration to measure reuse instead of tailoring rules to every test case. Track mapping/rule/code edits, regression tests and engineering time, including commissioning and correction. An accuracy tie can coexist with maintenance differences, but those differences must be measured.

D. Measurements, incident diagnosis and uncertainty

Primary outcomes are grounded-answer/evidence correctness, invalid proposals admitted, legal proposals rejected and complete legal task execution. Distinguish proposal errors, gateway rejection, controller rejection and physical outcome; successful PLC protection must not be credited as correct agent reasoning. Record retrieval, validation, dispatch, controller-acknowledgement and total latency separately, including p50/p95 and supervisory deadline misses. Freshness limits are selected for the workload, not copied from the benchmark's synthetic 15-minute cutoff. Use monotonic local durations and synchronised UTC for cross-device correlation, with clock uncertainty recorded.

For incident diagnosis, candidate causes and multi-cause labels are fixed before ranking. Vessel absence, an approved sensor outage or downstream blockage can provide interpretable episodes. A graph path or retrieved alarm is not a causal proof. MAP@3 is appropriate only for the defined ranked-cause task; the suggested 94% comparison cannot be used without matching inputs, dataset and protocol. Specific CausalTrace reproduction remains dependent on source and artefact access.

Use episode/asset-level held-out splits, not paraphrases, and treat model repeats within episodes as correlated. Pilot variability and engineer-agreed thresholds determine sample sizes. Even zero events in 300 independent trials gives an approximate 95% upper risk bound of 1%, not proven safety; correlated trials require episode-level intervals or clustered analysis. The supplied JSON schema records provenance, snapshot, proposal, approval, gateway/PLC decisions and actual outcome. Physical, hardware-in-the-loop and replay traces remain separate.

X. Limitations and Deployment Implications

The benchmark has one reference-plant model, authored tasks and injected faults. Equal-information procedural validation shares the reference rules and does not provide independent truth. Repeated model runs must be analysed within episodes, not treated as independent incidents. Corrupted constraints, incorrect mappings, incomplete interlock inventories, unseen fault combinations and future-dated evidence remain additional threats. The current harness is not a live dispatcher.

Physical evaluation must distinguish agent proposal quality from gateway rejection and PLC protection. Zero observed invalid actions cannot establish a safety integrity level. Dataspace permission is a separate untested extension: ODRL expresses policies, while trusted enforcement and downstream-use control require additional mechanisms; it is not automatically interchangeable with SHACL.

XI. Conclusion

The controlled corpus supports engineering-evidence checks, not an inherent accuracy advantage of semantic graphs over an equally informed rule program. The proposed framework is useful only insofar as it helps expose, select and maintain the facts industrial decisions require. Physical performance, independent capability assessment and transfer effort are the next evidence gates.

References

[1] International Electrotechnical Commission, "IEC 62541: OPC unified architecture," IEC, Tech. Rep., 2020, multi-part standard; information model in Part 3, address space model in Part 5.

[2] Industrial Digital Twin Association, "Specification of the asset administration shell — part 1: Metamodel; part 2: Application programming interfaces; part 3a: Data specification IEC 61360; part 5: Package file format (AASX)," IDTA, Tech. Rep., 2023. [Online]. Available: https://industrialdigitaltwin.org/

[3] D. Edge, H. Trinh, N. Cheng, J. Bradley, A. Chao, A. Mody, S. Truitt, and J. Larson, "From local to global: A graph RAG approach to query-focused summarization," arXiv:2404.16130, 2024. [Online]. Available: https://arxiv.org/abs/2404.16130

[4] S. Pan, L. Luo, Y. Wang, C. Chen, J. Wang, and X. Wu, "Unifying large language models and knowledge graphs: A roadmap," *IEEE Transactions on Knowledge and Data Engineering*, 2024.

[5] H. Knublauch and D. Kontokostas, "Shapes constraint language (SHACL)," W3C, W3C Recommendation, 2017. [Online]. Available: https://www.w3.org/TR/shacl/

[6] A. Todkar, M. Sarkar, and J. Solanki, "Semantic framework for IT-OT integration in industrial environments," *Annual Conference of the PHM Society*, vol. 17, no. 1, 2025. [Online]. Available: https://www.papers.phmsociety.org/index.php/phmconf/article/view/4551

[7] ECLASS e.V., "ECLASS — the cross-industry product data standard," ECLASS e.V., Tech. Rep., 2024. [Online]. Available: https://eclass.eu/

[8] International Electrotechnical Commission, "IEC 61360: Standard data element types with associated classification scheme," IEC, Tech. Rep., 2017, basis of the IEC Common Data Dictionary (CDD).

[9] ——, "IEC 61987: Industrial-process measurement and control — data structures and elements in process equipment catalogues," IEC, Tech. Rep., 2019.

[10] International Organization for Standardization, "ISO 29002-5: Industrial automation systems and integration — exchange of characteristic data — part 5: Identification scheme," ISO, Tech. Rep., 2010.

[11] International Electrotechnical Commission, "IEC 62714: Engineering data exchange format for use in industrial automation systems engineering — AutomationML," IEC, Tech. Rep., 2018.

[12] ——, "IEC 62264: Enterprise-control system integration (ISA-95)," IEC, Tech. Rep., 2013.

[13] VDI/VDE/NAMUR, "VDI/VDE/NAMUR 2658: Automation engineering of modular systems in the process industry — module type package," VDI, Tech. Rep., 2019.

[14] International Society of Automation, "ANSI/ISA-88 batch control and ANSI/ISA-TR88.00.02 (PackML) machine and unit states," ISA, Tech. Rep., 2008.

[15] W3C Web of Things Working Group, "Web of things (WoT) thing description 1.1," W3C, W3C Recommendation, 2023. [Online]. Available: https://www.w3.org/TR/wot-thing-description11/

[16] ETSI, "ETSI GS CIM 009: Context information management (CIM); NGSI-LD API," ETSI, Tech. Rep., 2023.

[17] International Electrotechnical Commission, "IEC 61850: Communication networks and systems for power utility automation," IEC, Tech. Rep., 2020.

[18] QUDT.org, "QUDT — quantities, units, dimensions and data types ontologies," https://qudt.org/, 2024.

[19] A. Haller, K. Janowicz, S. J. D. Cox, M. Lefrançois, K. Taylor, D. Le Phuoc, J. Lieberman, R. García-Castro, R. Atkinson, and C. Stadler, "The modular SSN ontology: A joint W3C and OGC standard specifying the semantics of sensors, observations, sampling and actuation," *Semantic Web*, vol. 10, no. 1, pp. 9–32, 2019.

[20] B. Balaji, A. Bhattacharya, G. Fierro, J. Gao, J. Gluck, D. Hong, A. Johansen, J. Koh, J. Ploennigs, Y. Agarwal, M. Berges, D. Culler, R. Gupta, M. B. Kjærgaard, M. Srivastava, and K. Whitehouse, "Brick: Towards a unified metadata schema for buildings," in *Proceedings of the 3rd ACM International Conference on Systems for Energy-Efficient Built Environments (BuildSys)*, 2016.

[21] International Data Spaces Association, "IDS reference architecture model, version 4," IDSA, Tech. Rep., 2022. [Online]. Available: https://internationaldataspaces.org/

[22] R. Iannella and S. Villata, "ODRL information model 2.2," W3C, W3C Recommendation, 2018. [Online]. Available: https://www.w3.org/TR/odrl-model/

[23] *First steps: Industrial Information Hub*, Industrial Operations X Documentation, Siemens, version 2.6 documentation. URL visited Oct. 8, 2026. [Online]. Available: https://docs.industrial-operations-x.siemens.cloud/r/en-us/v2.6/industrial-information-hub/first-steps